ZEAL COLLEGE OF ENGINEERING & RESEARCH, PUNE
PRESENTS

AlignX

Offical Event Rulebook

AVISHKAR '26
WHERE CURIOSITY BECOMES CREATION

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1. Event Overview

- Event Name: Line Follower Challenge
- Event Type: Part of Avishkar 2K26 – ZCOER
- Mode: Offline
- Team Size: Team of 4 Members
- Bot Type: Autonomous Robot

2. Event Task / Objective

- Purpose: Foster practical understanding of robotics, sensor integration, and autonomous systems while encouraging innovation and real-time problem solving.
- Task: Participants must build a fully autonomous robot to follow a thick black line on a white surface, navigating sharp turns, curves, intersections, and crossovers without manual intervention.

3. Design / Technical Specifications

- Autonomy: Robots must be completely autonomous and wireless (no Bluetooth, RF, or WiFi).
- Physical Limits: Maximum weight of 3kg (5% tolerance) and must fit within a 20cm x 20cm x 20cm box.
- Power: Maximum voltage must not exceed 16.8V DC.
- Required Features: Must include a fail-safe mechanism and support at least two autonomous speed modes ("Precision" and "Turbo").
- Efficiency: Teams must declare battery capacity; bonus points are awarded for maximum distance covered per mAh consumed.

4. General Guidelines

- Track Details: The track features a 3cm thick black line on a white background, including hairpin bends, acute angles, and potential line color interchanges (white line on black).
- Prohibitions: Shortcuts or skipping sections are strictly prohibited.
- Preparation: Each team is allowed 15 minutes for robot calibration before each round.
- Track Reveal: The actual track layout will only be revealed on the day of the event.

5. Safety Rules

- Assembly: All components, including batteries and wiring, must be securely assembled.
- Prohibited Materials: No flammable, hazardous, or high-voltage materials are allowed.
- Venue Conduct: Participants must follow all venue safety instructions and spectators must stay outside operational areas.

6. Event Rounds / Structure

Round	Description	Duration
Round 1	Preliminary (Timed runs on standard track)	**3 hours**
Round 2	Finals (Top 10 teams compete)	2 hours
Final Evaluation	Judging & Prize Distribution	1 hour

7. Submission Guidelines

Final submissions must include:

- A working robot
- Source code and schematic

8.Judging Criteria:

Criteria	Weightage
Accuracy & Precision	25%
Adaptability	25%
Speed and Efficiency	20%
Design and Construction	15%
Technical Implementation	15%

9. Scoring Overview

Parameter	Max Points
Performance / Accuracy	30
Speed & Efficiency	25
Adaptability	20
Design & Construction	15
Code & Innovation	10
Total	**100**

10. Penalties & Disqualifications

Violation	Penalty
Exceeding weight/size limits	Disqualification
Wireless communications	Disqualification
Unsafe Operation	Immediate Disqualification
Manual intervention during the run	-30 Points
Shortcuts/ Skipping sections	-30 Points/Disqualification
Violation of Safety Zone	-20 Points
Incomplete Work	-10 Points
Misconduct or Unethical Practice	Permanent Ban

11. Event Team

- Event Heads (Robotics & Automation):
  - Mohit Meshram: 8692863047
  - Aaryan Nikam: 8412985636
  - Sanyog Dahotre: 8007300014

12. General Instructions

- Respect must be shown to all judges, peers, and staff.
- Plagiarism or code reuse will result in disqualification.
- All decisions made by the judges are final and binding.